Network Hardening Lab

April 24, 2026

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Purpose of Project

- Cybersecurity graduates should have experience in network building and securing.
  - Helps to find problems in networks
  - Helps to maintain networks
- Solution: Provide a lab to build and secure a practice range network
  - Gives experience
  - Helps connect concepts to hands on experience

Research

- Mostly used knowledge and experience from a past class
- Looked up YouTube videos for devices I didn’t know how to set up
  - ASA / Router / Firewall
  - Wireless Access Point

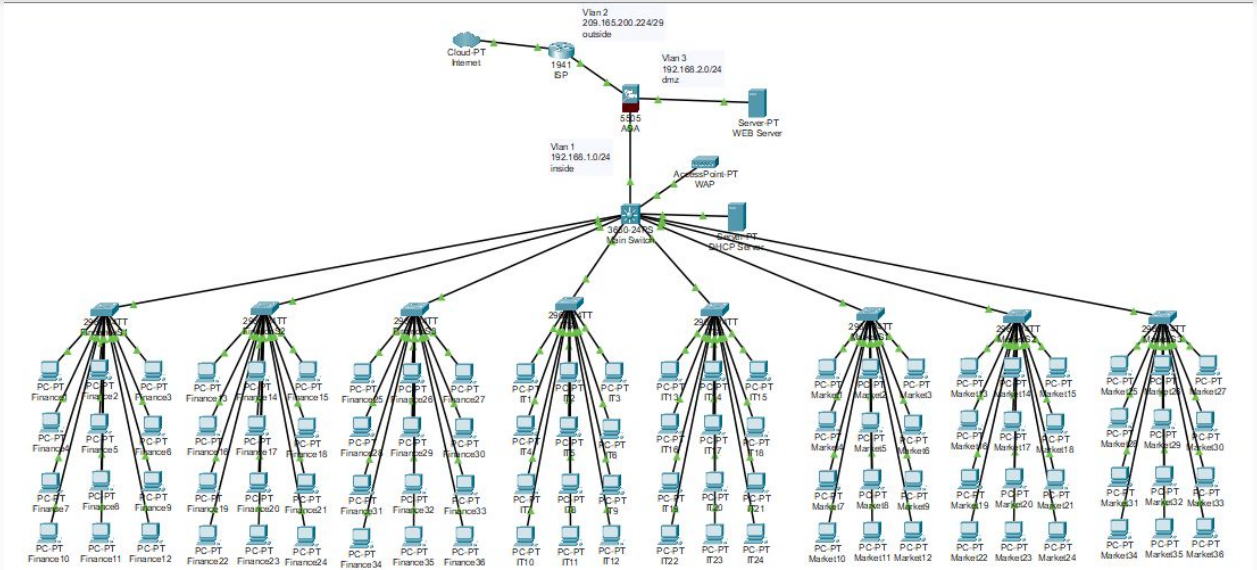
Research Reference Links

- ASA Config Part 1: https://www.youtube.com/watch?v=SLZS1mSc_VY&t=1037s
- ASA Config Part 2: https://www.youtube.com/watch?v=uJfI69mGUeU
- ASA Config Part 3: https://www.youtube.com/watch?v=IQI-Rv8cmYM&t=4s
- ASA Config Part 4: https://www.youtube.com/watch?v=pBW1X6r5kNM&t=845s
- Access Point configuration: https://www.youtube.com/watch?v=IQakg45Fe5Q

Software and Hardware

- Cisco Packet Tracer application
  - Free
  - Easy to use
  - Simulates hardware
  - Low computer resource requirements
  - Slight limitations to end point devices

Project Requirements



Students will connect and secure a Router/Firewall, DHCP server, Wireless Access Point, WEB server, several switches, and ~100 PCs.

Demonstration

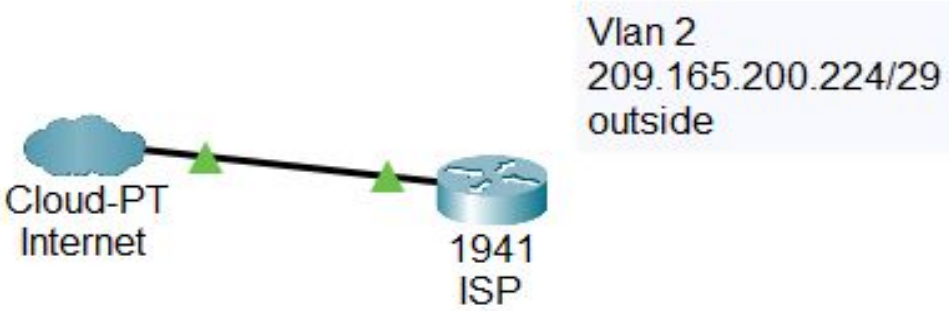


Fig 1. The initial outside network for the user to build off of

Demonstration

The lab will lead the user through making a network and base the hardening of the network based on the security plan that is provided with the lab:

Security Plan

The security plan describes the hardening requirements of Routers, Switches, Firewalls, and more networking devices, as well as the hardening requirements for endpoint devices, even the ones that CISCO Packet Tracer doesn’t allow.

Demonstration

Network Security

- Configure a network firewall with ACLs blocking unauthorized outside network traffic
- Configure the network to be separated into VLANs to keep network traffic separated
- Enable passwords in configurations for devices that can have passwords set

Endpoint Security

- Put passwords on endpoint devices that will let you
- Ensure encryption is used in traffic from devices

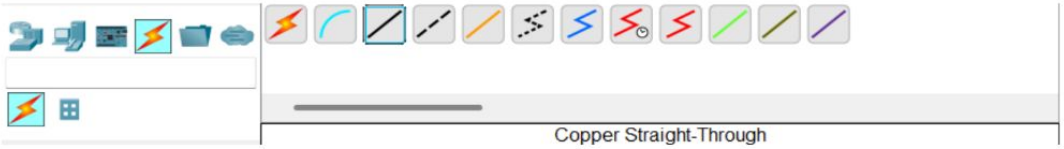
Fig 2. Snippets of Network Device and Endpoint Device Security Requirements

Demonstration

The lab contains detailed instructions to make it it easier to follow including color coded instructions and pictures.

```
ciscoasa# conf t
        Entering configuration mode
ciscoasa(config)# hostname MainASA
        Renaming the terminal
MainASA(config)# enable password Cisco123
```

Fig 3. Snippet of color coded instructions



Click on the ASA you put down with the wire connection and a list of connection points should appear. Click on Ethernet0/0 then click on the ISP router and connect the wire to GigabitEthernet0/1. This should connect a wire between the two devices:

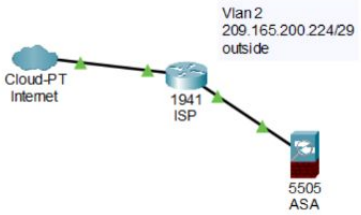


Fig 4. Snippet of a section in the lab instructions

Demonstration

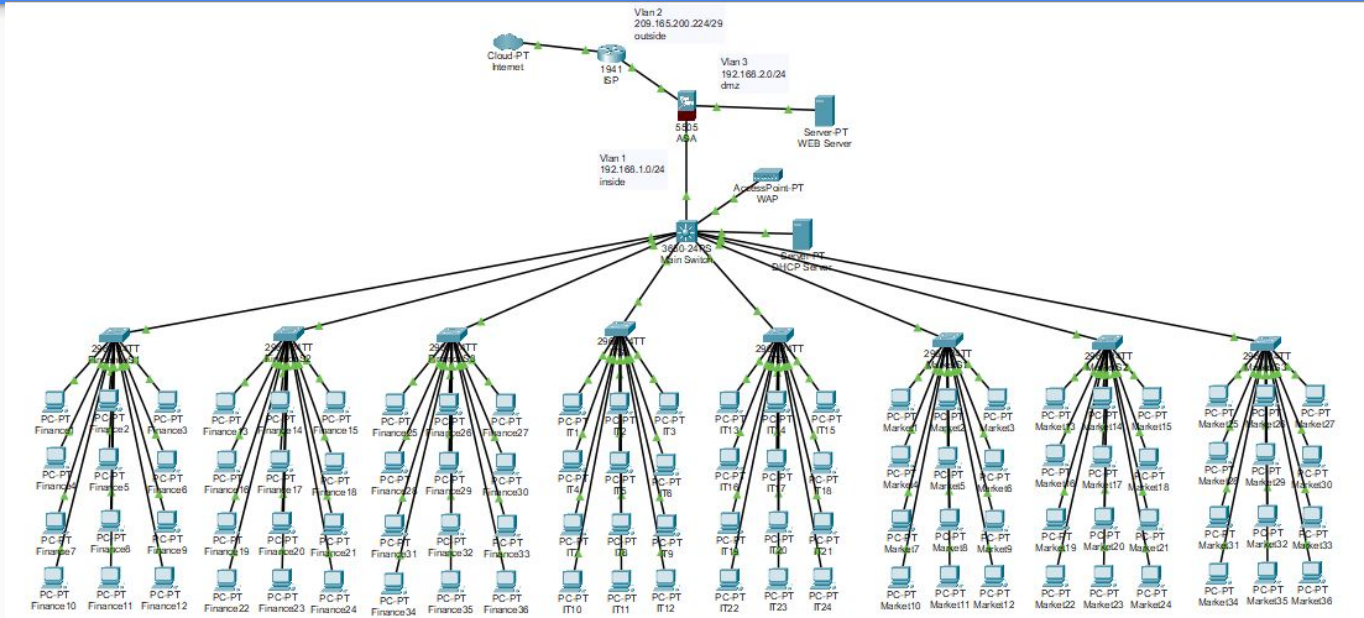


Fig 5. Finished network that is fully connected and hardened

Test Plan

- Tests that user can set up Cisco Packet Tracer
- Tests that network has full connectivity with authorized devices
  - Tests physical and wireless devices separately
- Tests that network can’t connect with unauthorized devices
- User Testing: Provided a link to the CSU cybersecurity Discord and contacted some friends
- Test Plan

Test Results

- Cisco Packet Tracer is easy to connect and set up.
- Connectivity throughout the network is good and only authorized outside devices are able to be connected to.
- WEB server is able to be pinged but could not get the WEB itself to work through the URL
- Recieved feedback from only two users:
  - Instructions were clear and informative
  - Good for less experienced Cybersecurity specialists
  - Easy to set up
  - Generally good feedback

Challenges

- Limitations with Cisco Packet Tracer
  - Forced to cut out scripts
  - Not as much security options for endpoint security
  - Network devices oriented
  - Worked around by explaining the concepts in the lab
- Issues with receiving user feedback
  - Takes a long time
  - Poor distribution methods
  - Only got a couple feedbacks from friends

Future Enhancements

- New future devices may need to be implemented
- Possibly better applications to use other than Cisco Packet Tracer
- Implement a “second building” which adds new concepts
- Configure better or more security settings as my knowledge grows

Questions